

User-Guided Variable-Rate Time-Stretching Via Stiffness Control



Nicholas J. Bryan, Jorge Herrera, and Ge Wang

Stanford University | CCRMA

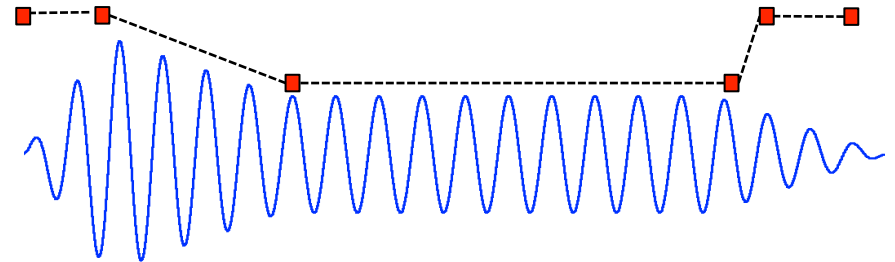
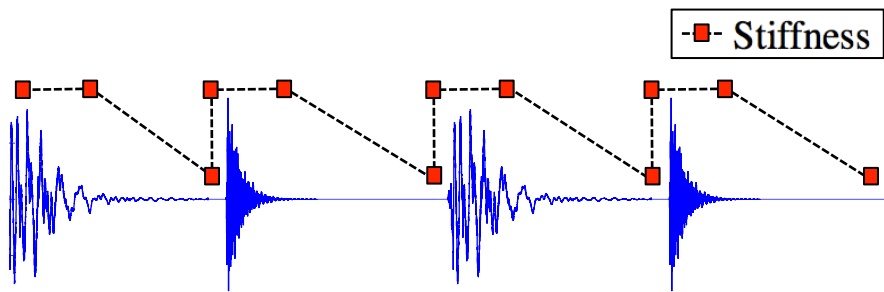
CCRMA DSP Seminar, November 6th 2012

Outline

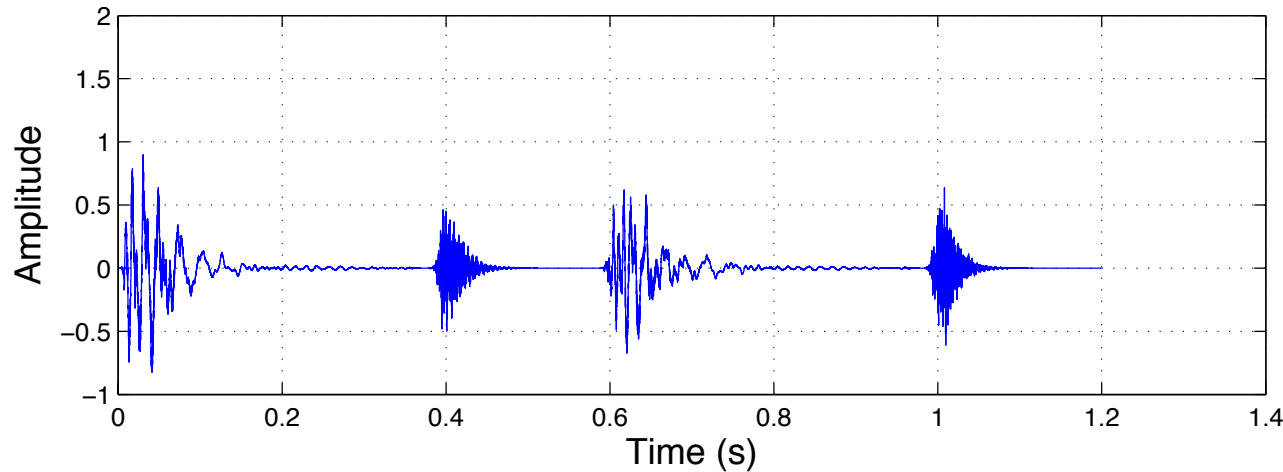
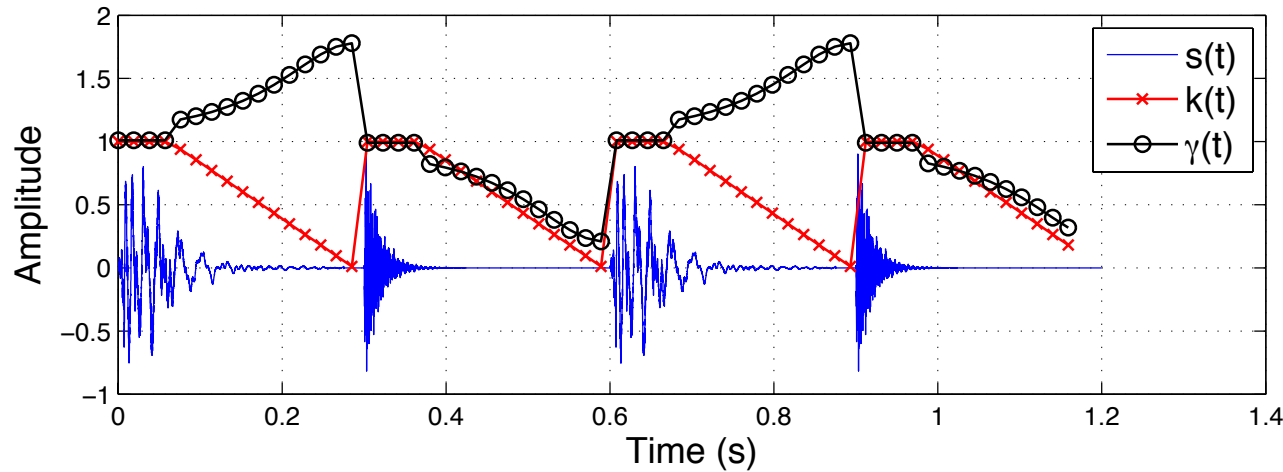
- I Introduction
- II Proposed Method
- III Numerical Optimization
- IV Time-Stretch Processor
- V Results
- VI Conclusions

Introduction

- User control over variable-rate time-stretch processing
- Stretch some regions more than others (e.g. stretchability, stiffness)
- Transient preservation, rhythmic warping, emphasis modification, etc.



Demo - Rhythmic Warping



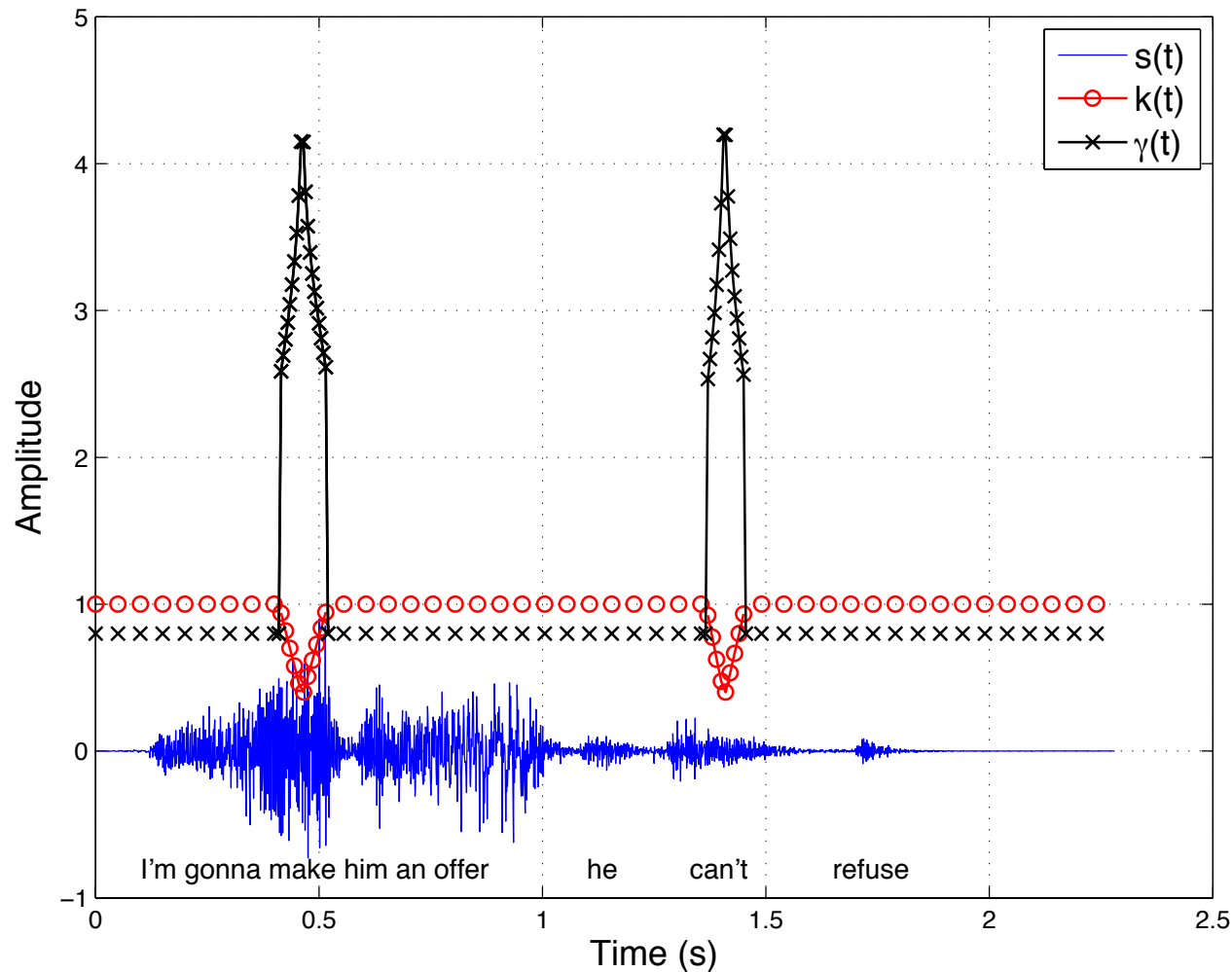
 Original

No Stiffness / Stiffness

 Stretched (2x) 

 Swung +
Stretched (2x) 

Demo - Emphasis Modification



 original

 warped

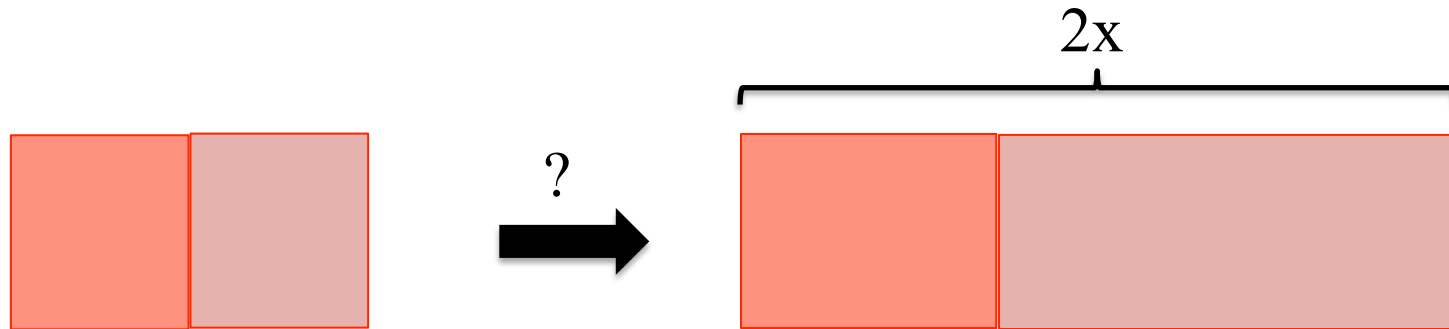
 warped + stretched
(slowed by 1.3x)

“I’m gonna make him an offer he can’t refuse.”

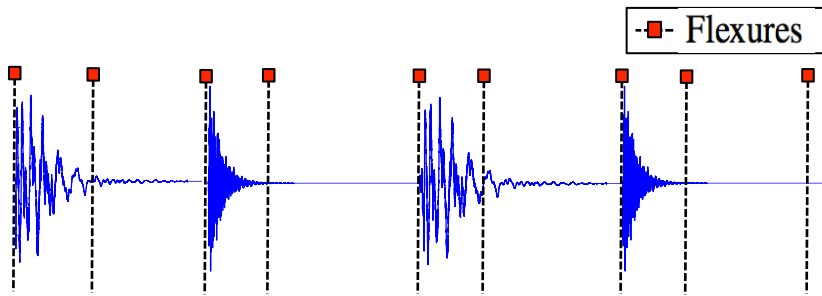
“I’m gonna maaake him an offer he caaaan’t refuse.”

Motivation

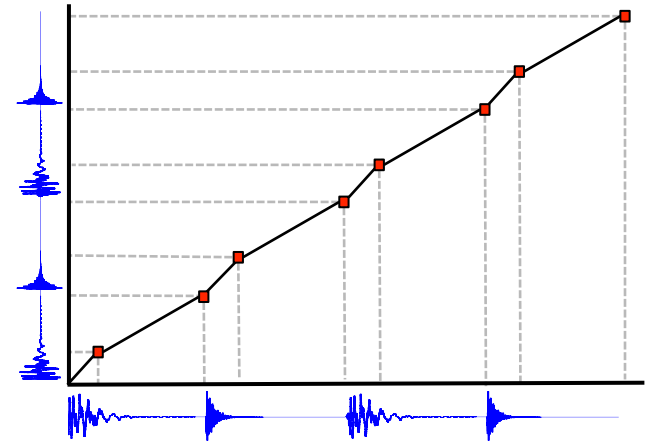
- **Automatic** methods have no mechanism for user input
- **Direct** manipulation of the stretch rate is hard!



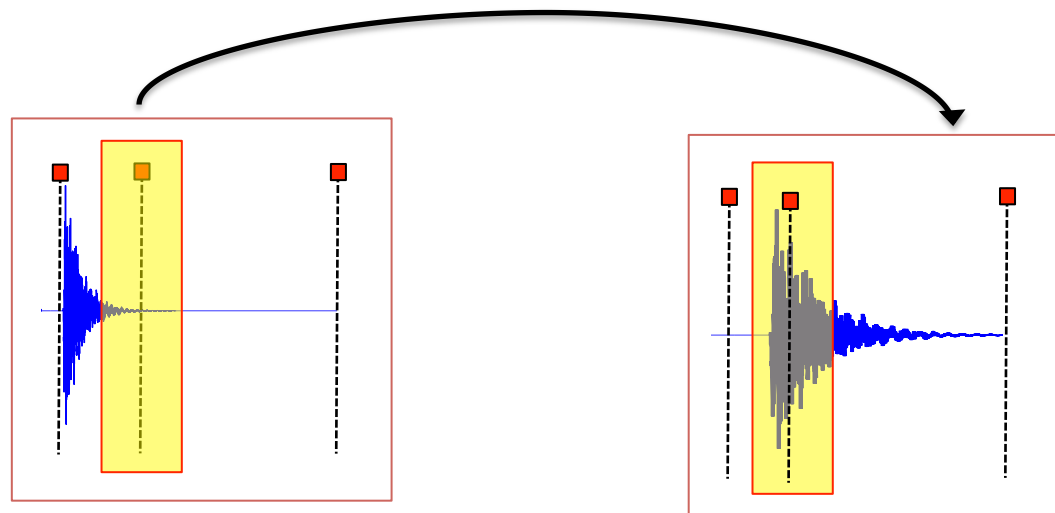
Flexures and Pivots



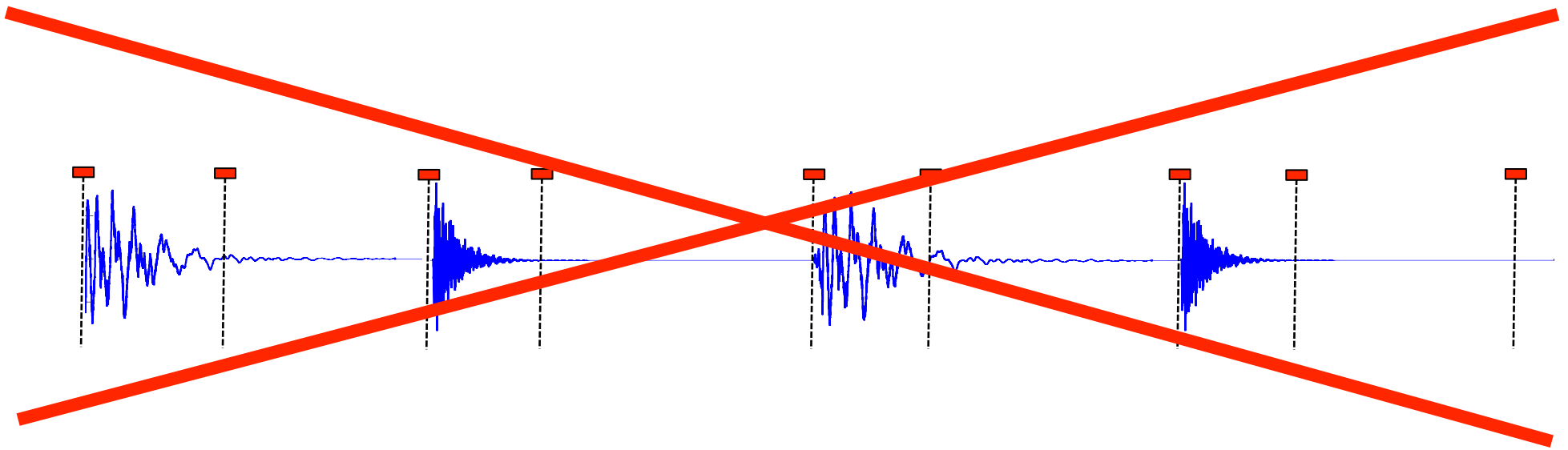
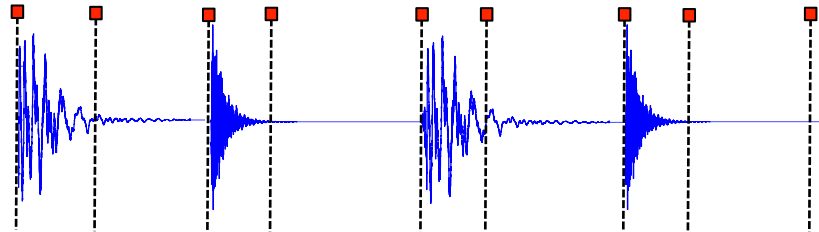
[ProTools, Logic Pro, FL Studio, etc.]



[Nielson and Brandorff, 2002]



Coupled Constraints

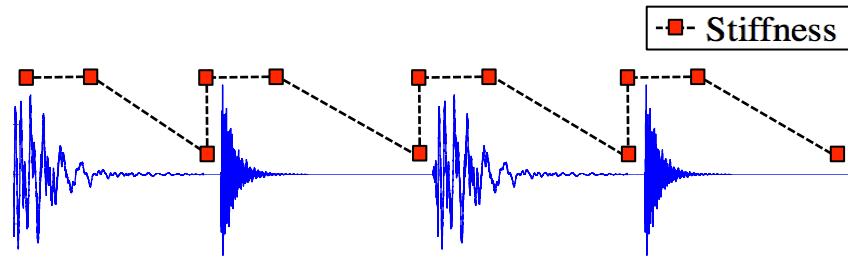


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Proposed Method

1. User annotates **stiffness** (+ timing **constraints**)

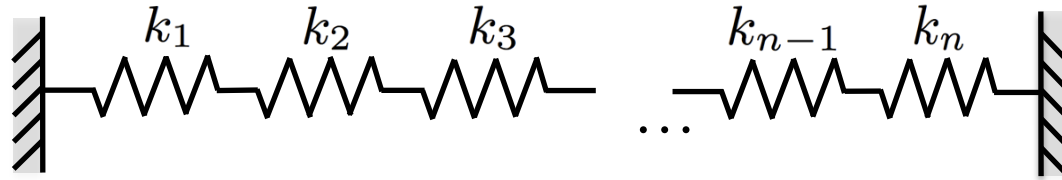


2. Solve optimization problem to **convert** stiffness to stretch factor
3. Use **pre-existing** time-stretch processor to stretch sound

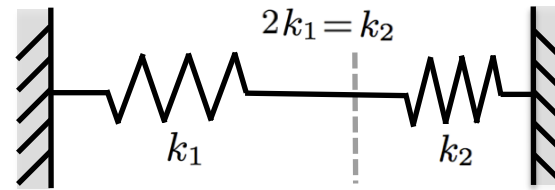
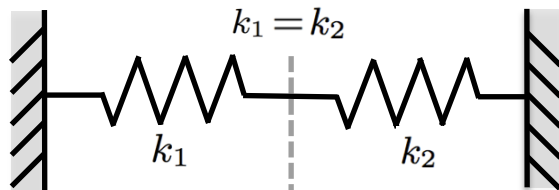


Step I: Spring Chain

- Model audio as **chain of springs**



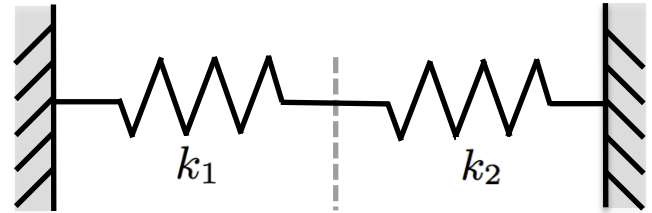
- Solve for **equilibrium** via Hooke's Law $F_i = -k_i x_i$



- Spring stiffness offers an **intuitive** measure (i.e. proportional)

Initial Formulation

$$\begin{array}{ll}\text{find} & \mathbf{x} \\ \text{subject to} & \mathbf{f} = \mathbf{0} \\ & \mathbf{x}^T \mathbf{1} = L\end{array}$$



$\mathbf{x}$ = spring displacement

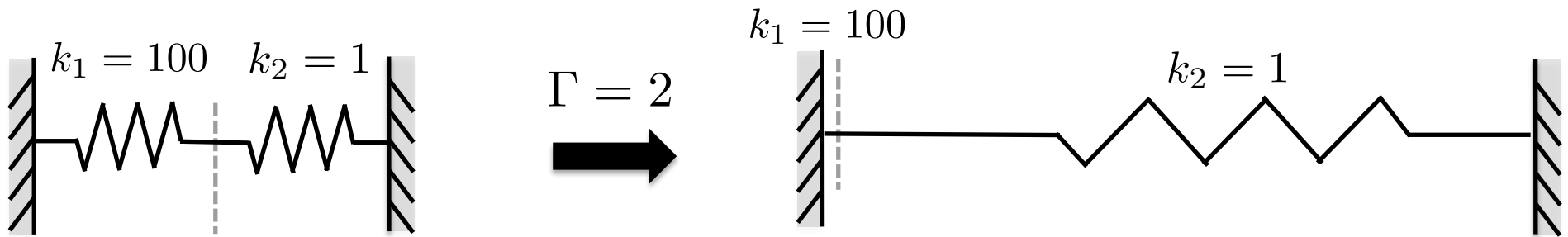
$\mathbf{f}$ = spring forces $f_i = k_{i+1}x_{i+1} - k_i x_i$

$\mathbf{k}$ = spring stiffness

L = final length

Problem

- Violates intuition: no initial rest length



Reformulation

$$\begin{array}{ll} \text{find} & \mathbf{x} \\ \text{subject to} & \mathbf{f} = \mathbf{0} \\ & \mathbf{x}^T \mathbf{1} = L \end{array} \longrightarrow \begin{array}{ll} \text{find} & \mathbf{x} \\ \text{subject to} & \mathbf{f} = \mathbf{0} \\ & (\mathbf{x} + \mathbf{x}_0)^T \mathbf{1} = L \\ & \mathbf{x} + \mathbf{x}_0 \geq \mathbf{0} \end{array}$$

$\mathbf{x}_0$ = Initial Rest Length

Reformulation

$$\begin{aligned} & \underset{\boldsymbol{x}}{\text{minimize}} && ||\boldsymbol{f}||_2 + \mu ||\boldsymbol{x}||_2 \\ & \text{subject to} && (\boldsymbol{x} + \boldsymbol{x}_0)^T \mathbf{1} = L \\ & && \boldsymbol{x} + \boldsymbol{x}_0 \geq \mathbf{0} \end{aligned}$$

$\boldsymbol{x}_0$ = Initial Rest Length

μ = Penalty Weight

- Minimize the force disturbance from equilibrium (smoothly)
- Minimize $\approx$ potential energy

Extensions

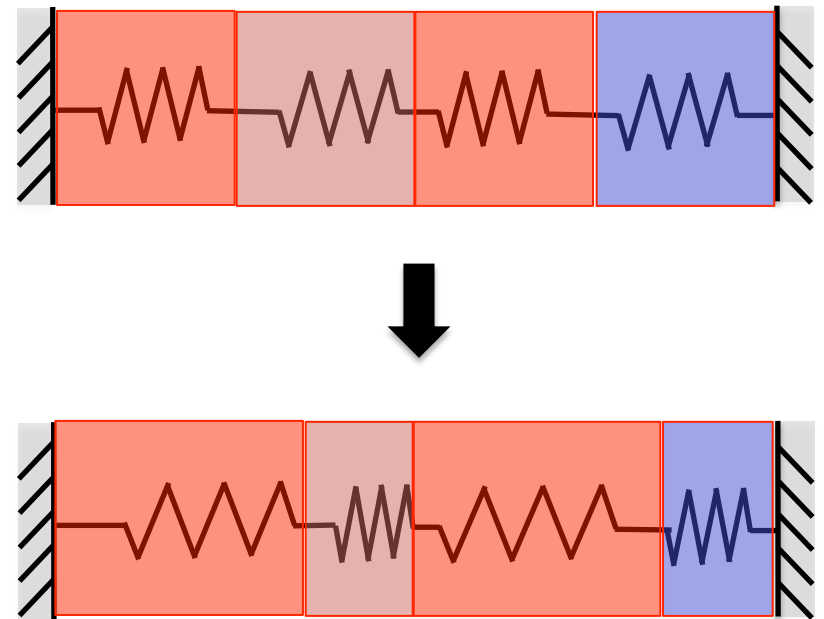
- Rhythmic warping, smoothing of user input, max stretching limits
- Example: Straight-to-Swing

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && ||\mathbf{f}||_2 + \mu ||\mathbf{x}||_2 \\ & \text{subject to} && (\mathbf{x} + \mathbf{x}_0)^T \mathbf{1} = L \\ & && \mathbf{x} + \mathbf{x}_0 \geq \mathbf{0} \end{aligned}$$

$$(x^1 + x_0^1)^T \mathbf{1} = \frac{2}{3} L/2$$

$$(x^2 + x_0^2)^T \mathbf{1} = \frac{1}{3} L/2$$

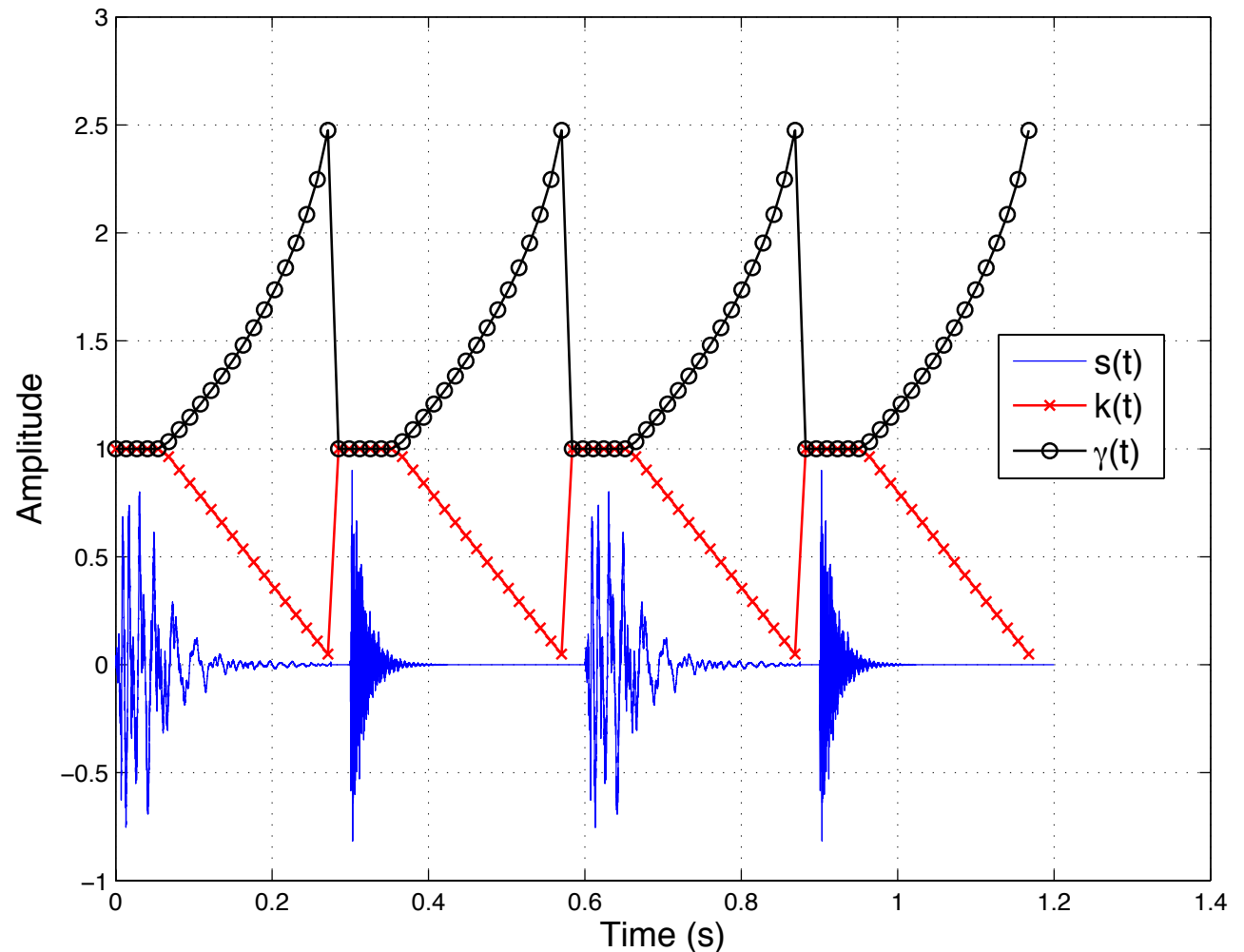
$$(x^3 + x_0^3)^T \mathbf{1} = \frac{2}{3} L/2$$



Step II: Stiffness to Stretch Factor

- Given input and output lengths, compute stretch factor as simple ratio

$$\gamma = \frac{\mathbf{x} + \mathbf{x}_0}{\mathbf{x}_0}$$



Step III: Time-Stretching

- Given optimal variable-rate stretch factor, process sound
- Phase Vocoder (PV)
- Pitch synchronous overlap add (PSOLA)

Method Recap

- Annotation method for user-annotated time stretching
- Decouples timing constraints and stiffness
- Solve optimization problem to get time-variable stretch-rate

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Numerical Computation I

- Solve the linearly constrained quadratic program

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \|\mathbf{f}\|_2 + \mu\|\mathbf{x}\|_2 \\ \text{subject to} & (\mathbf{x} + \mathbf{x}_0)^T \mathbf{1} = L \\ & \mathbf{x} + \mathbf{x}_0 \geq \mathbf{0} \end{array} \longrightarrow \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + \mathbf{r} \\ \text{subject to} & \mathbf{A} \mathbf{x} = \mathbf{b} \\ & \mathbf{C} \mathbf{x} \leq \mathbf{d} \end{array}$$

- Need iterative, numerical optimization method
- See CVX [Grant and Boyd, 2008] and [Boyd & Vandenberghe]

Descent-Based Methods

- Initialize x
- Repeat Until Convergence

{ • General Descent Methods

$$x^{n+1} = x^n + \tau^n \Delta x^n$$

- Gradient Descent

$$x^{n+1} = x^n - \tau^n \nabla f(x^n)$$

- Newton's Method

$$x^{n+1} = x^n - \tau^n \nabla^2 f(x^n)^{-1} \nabla f(x^n)$$

- Newton's Method w/Equality Constraints

$$x^{n+1} = x^n + \tau^n \Delta x^n$$

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

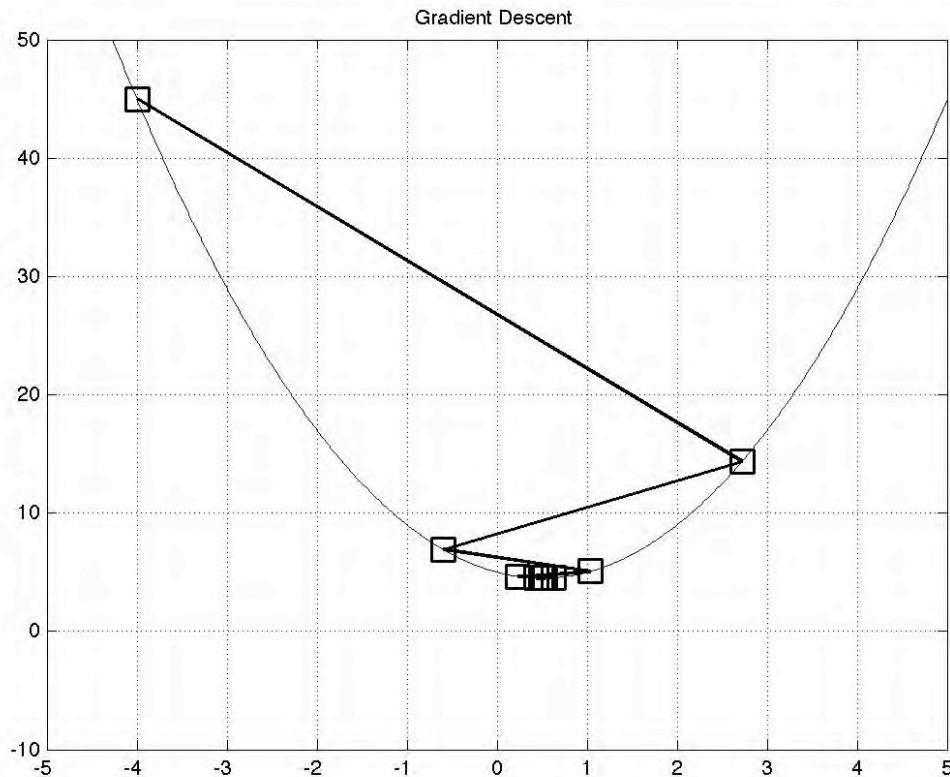
- Interior-Point Barrier Method (+inequality constraints)

}

Descent-Based Methods Graphically I

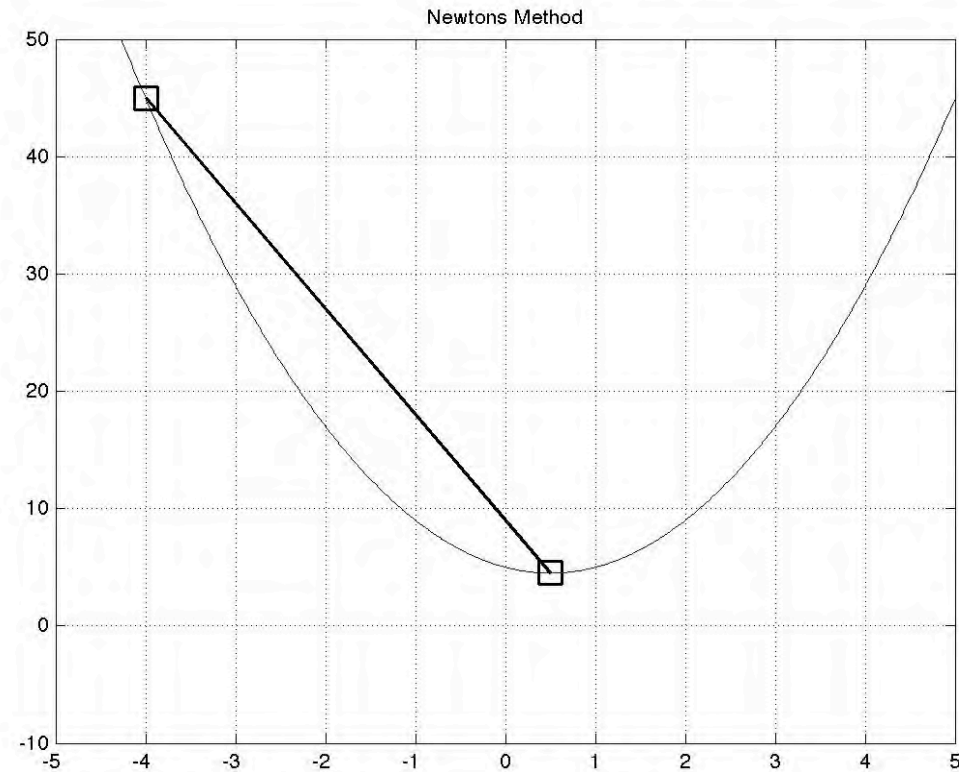
Gradient Descent

$$x^{n+1} = x^n - \tau^n \nabla f(x^n)$$



Newton's Method

$$x^{n+1} = x^n - \tau^n \nabla^2 f(x^n)^{-1} \nabla f(x^n)$$



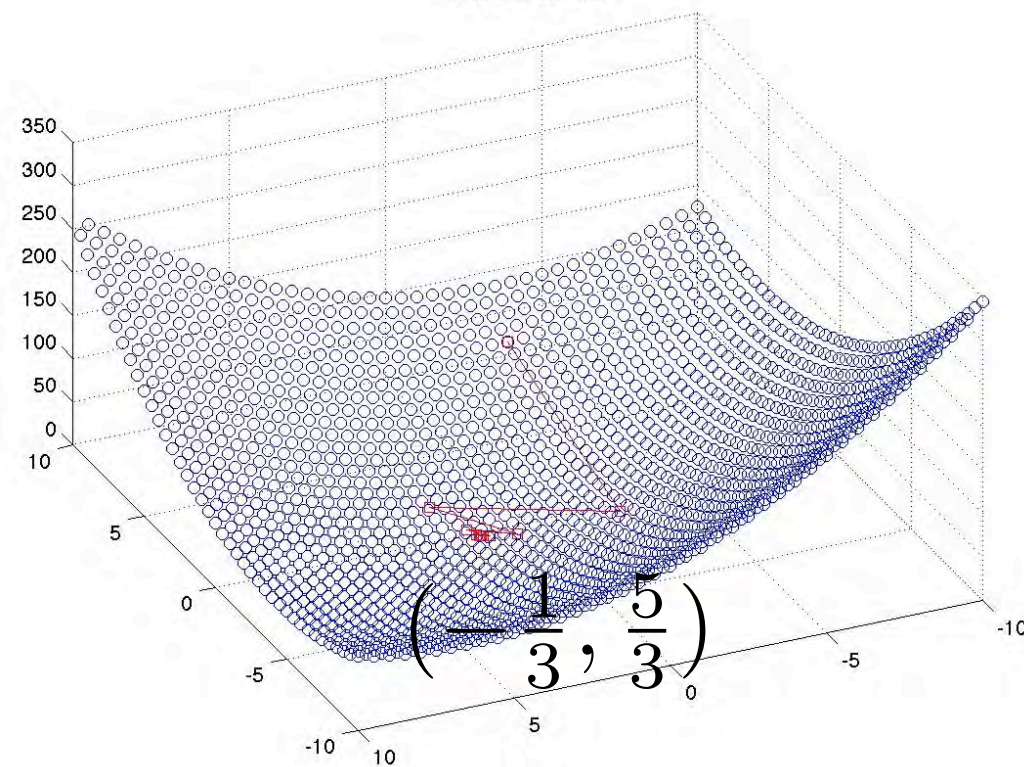
$$y = 2x^2 - 2x + 5$$

Descent-Based Methods Graphically II

Gradient Descent

$$x^{n+1} = x^n - \tau^n \nabla f(x^n)$$

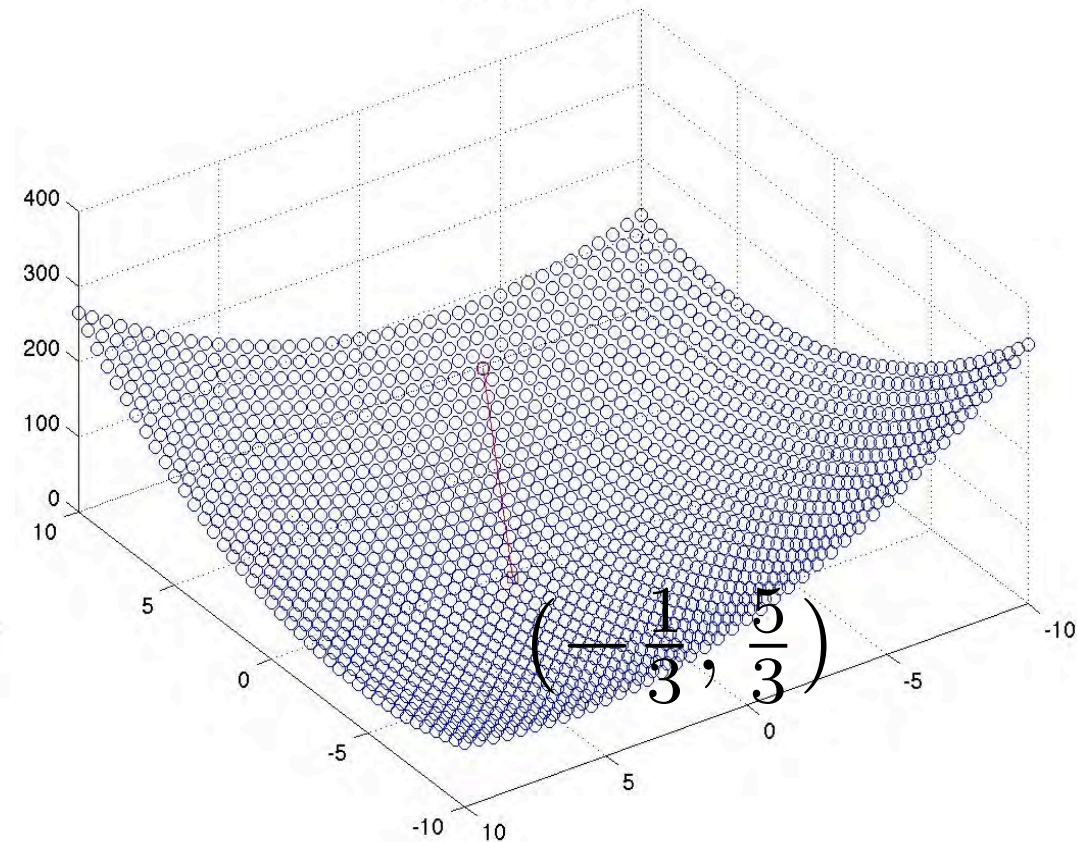
Gradient Method



Newton's Method

$$x^{n+1} = x^n - \tau^n \nabla^2 f(x^n)^{-1} \nabla f(x^n)$$

Newtons Method

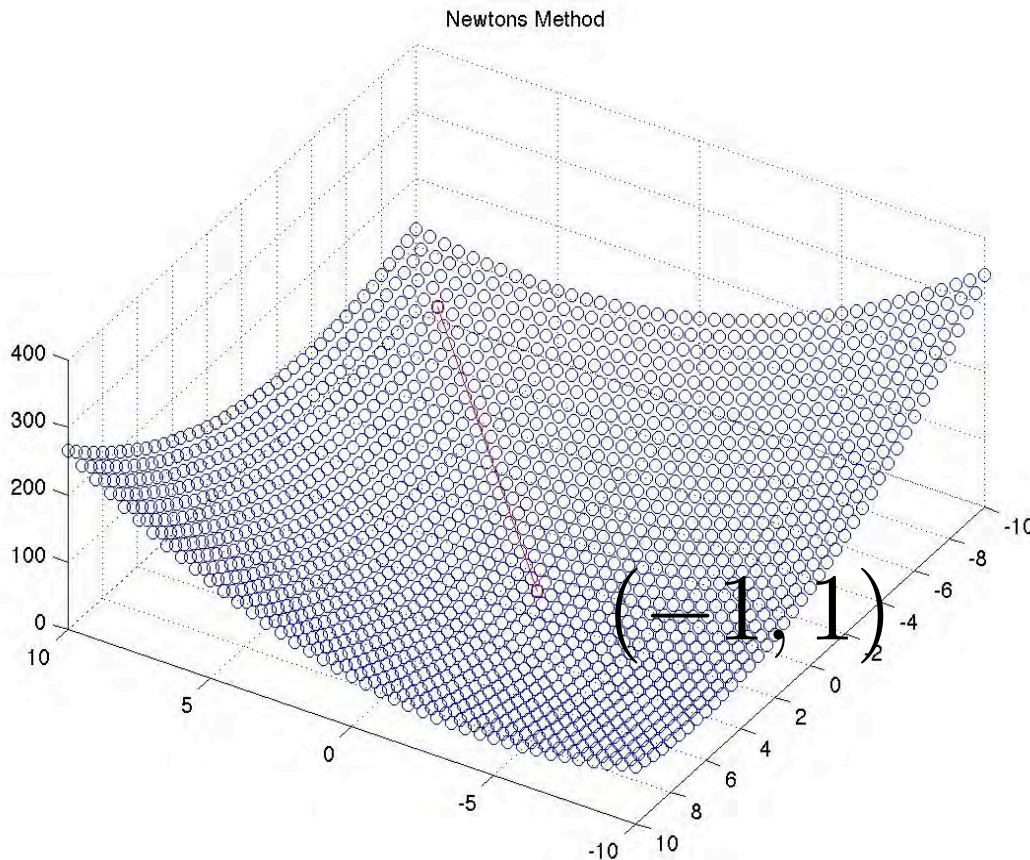


$$y = x^T \begin{pmatrix} 1 & .5 \\ .5 & 1 \end{pmatrix} x - \left(\frac{1}{3}\right)^T x + 5$$

Descent-Based Methods Graphically III

Newton's Method w/Equality Constraints

$$x^{n+1} = x^n + \tau^n \Delta x^n \quad \begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$



$$y = x^T \begin{pmatrix} 1 & .5 \\ .5 & 1 \end{pmatrix} x - \begin{pmatrix} 1 \\ 3 \end{pmatrix}^T x + 5$$

subject to $x_1 = -x_2$

Barrier Interior-Point (Newton) Method

- Move inequalities to main objective via log barrier

$$\begin{array}{ll}\underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T\mathbf{P}\mathbf{x} + \mathbf{q}^T\mathbf{x} + \mathbf{r} \\ \text{subject to} & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{C}\mathbf{x} \leq \mathbf{d}\end{array}$$



$$\begin{array}{ll}\underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T\mathbf{P}\mathbf{x} + \mathbf{q}^T\mathbf{x} + \mathbf{r} - (1/t)\mathbf{1}^T \log(-\mathbf{C}\mathbf{x} + \mathbf{d}) \\ \text{subject to} & \mathbf{A}\mathbf{x} = \mathbf{b}\end{array}$$

$$t > 0 \quad t \rightarrow \infty$$

Solve Sequential Optimizations

$$t = 1 \quad \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + \mathbf{r} - (1/t)\mathbf{1}^T \log(-\mathbf{C} \mathbf{x} + \mathbf{d}) \\ \text{subject to} & \mathbf{A} \mathbf{x} = \mathbf{b} \end{array}$$

$$t = 5 \quad \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + \mathbf{r} - (1/t)\mathbf{1}^T \log(-\mathbf{C} \mathbf{x} + \mathbf{d}) \\ \text{subject to} & \mathbf{A} \mathbf{x} = \mathbf{b} \end{array}$$

$$t = 10 \quad \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + \mathbf{r} - (1/t)\mathbf{1}^T \log(-\mathbf{C} \mathbf{x} + \mathbf{d}) \\ \text{subject to} & \mathbf{A} \mathbf{x} = \mathbf{b} \end{array}$$

$\vdots$

$\vdots$

$$t \rightarrow \infty \quad \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & (1/2)\mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + \mathbf{r} - (1/t)\mathbf{1}^T \log(-\mathbf{C} \mathbf{x} + \mathbf{d}) \\ \text{subject to} & \mathbf{A} \mathbf{x} = \mathbf{b} \end{array}$$

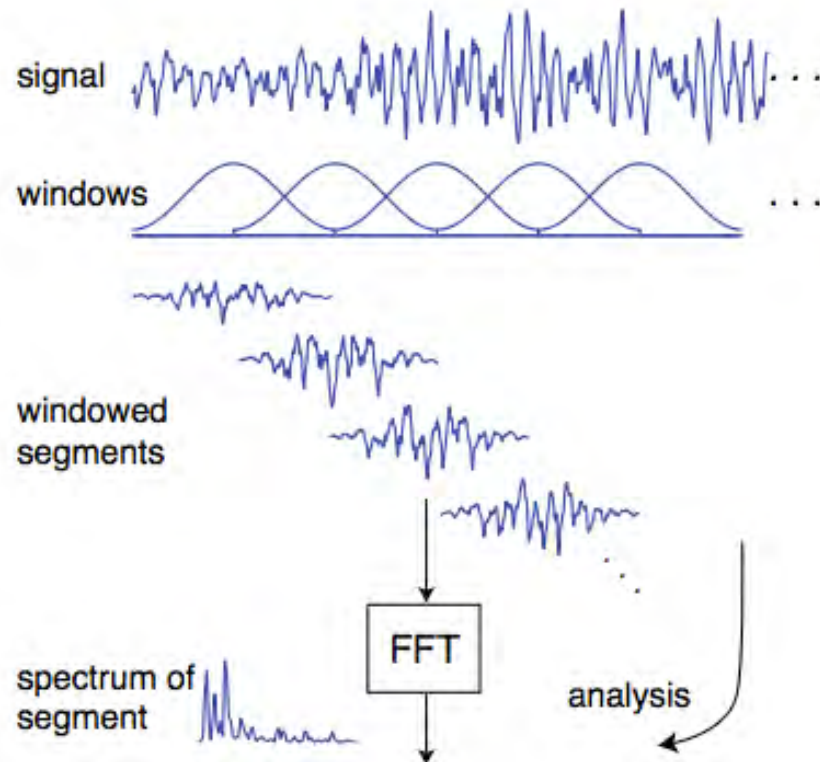
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Phase Vocoder

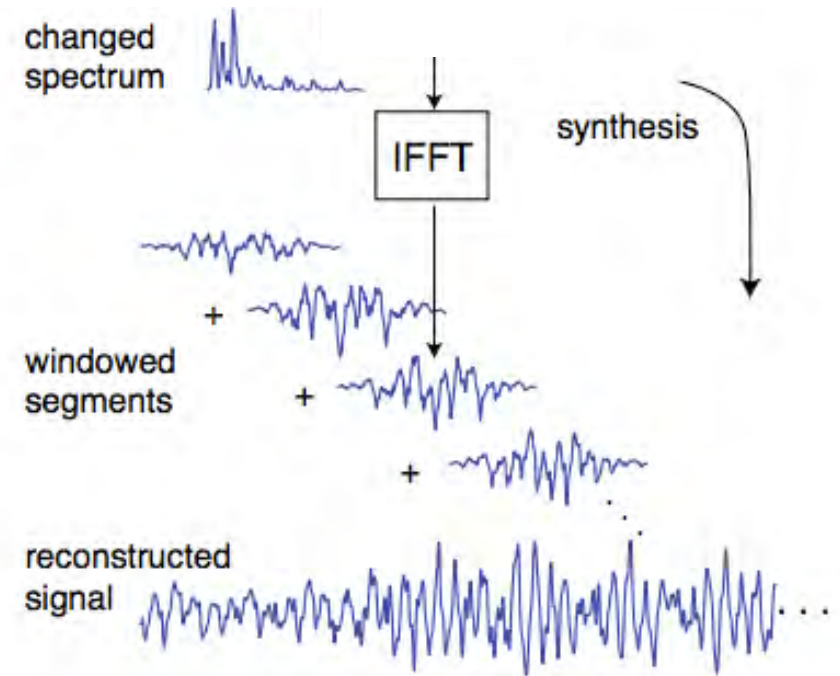
- Frequency domain method
 - Time-stretch
 - Pitch-shift
- General procedure
 - Filter Banks/STFT Analysis
 - Frequency domain processing
 - Synthesis
- Good for processing full-bandwidth audio signals

Phase Vocoder Graphically



(a) Analysis

spectral
manipulation



(c) Synthesis

Vocoder Time-Stretching

- Variable-rate time-stretching
 - Changing analysis hopsize, constant synthesis hopsize
- Linearly resample stretch-rate control signal as needed
- See websites for phase-vocoders in Matlab
 - [Dan Ellis 2002, Peter Moller-Nielsen 2002, William Sethares 2007]

Synchronous Overlap and Add (SOLA)

1. Segment signal into overlapping blocks (fixed length/step size)
2. Shift the blocks according to scale factor
3. Compute cross-correlation
4. Find where cross-correlation is maximum
5. Adjust the shift to synchronize blocks with maximum similarity
6. Cross-fade blocks

[Roucos *et al.*, 1985 and Makhoul *et al.*, 1986.]

SOLA Graphically

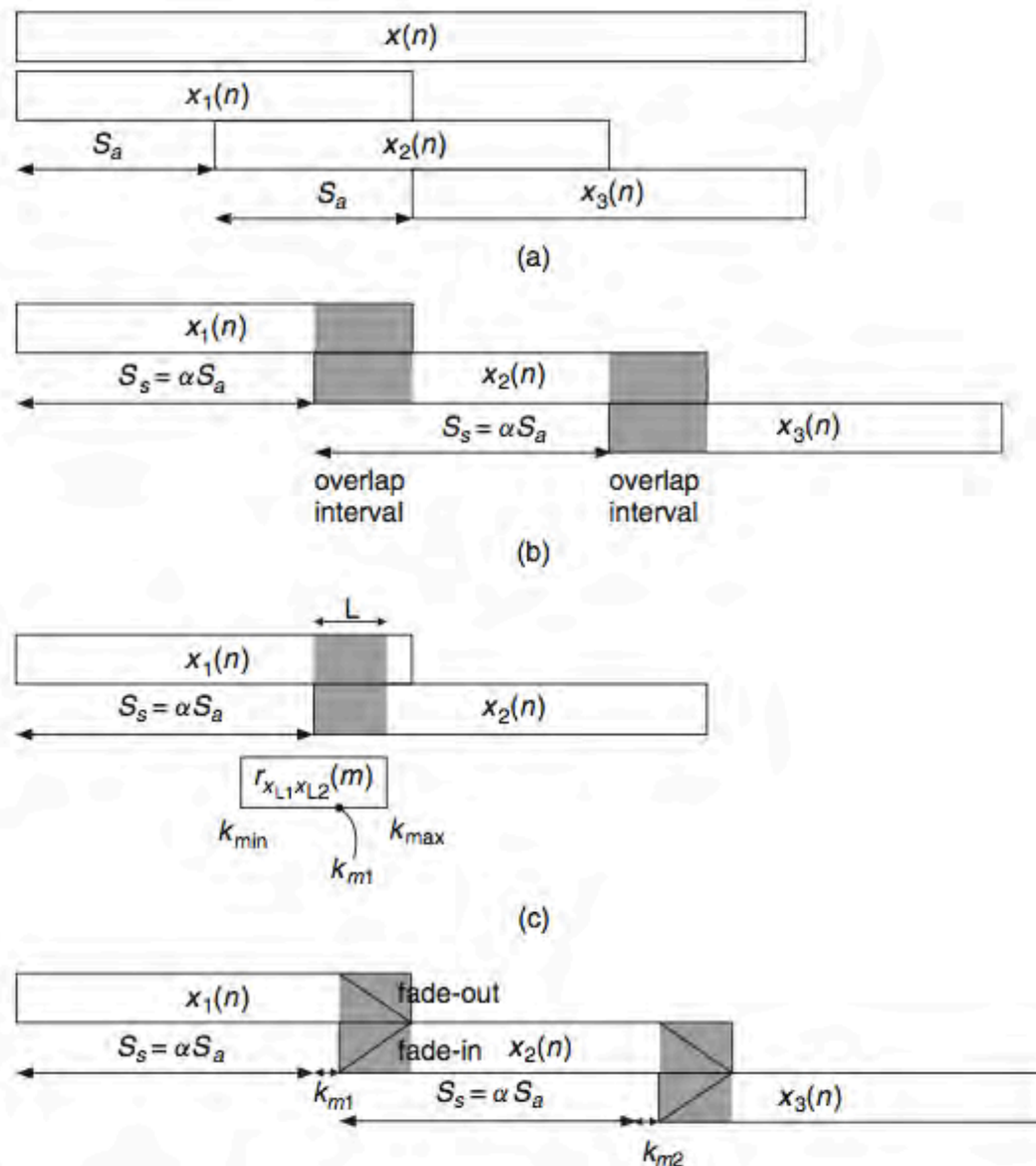
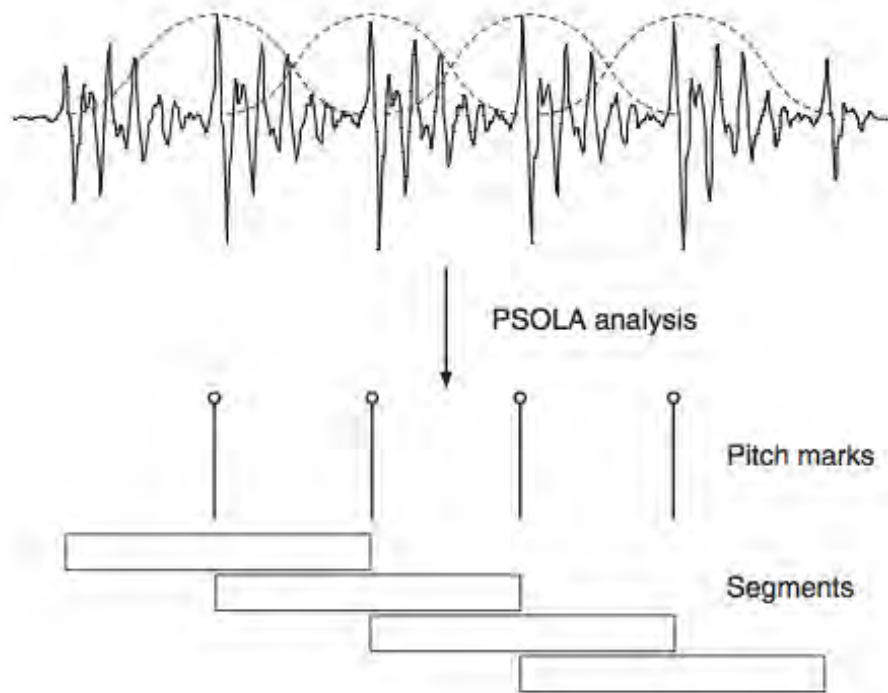


Figure from [Dutilleux et al. 2011]

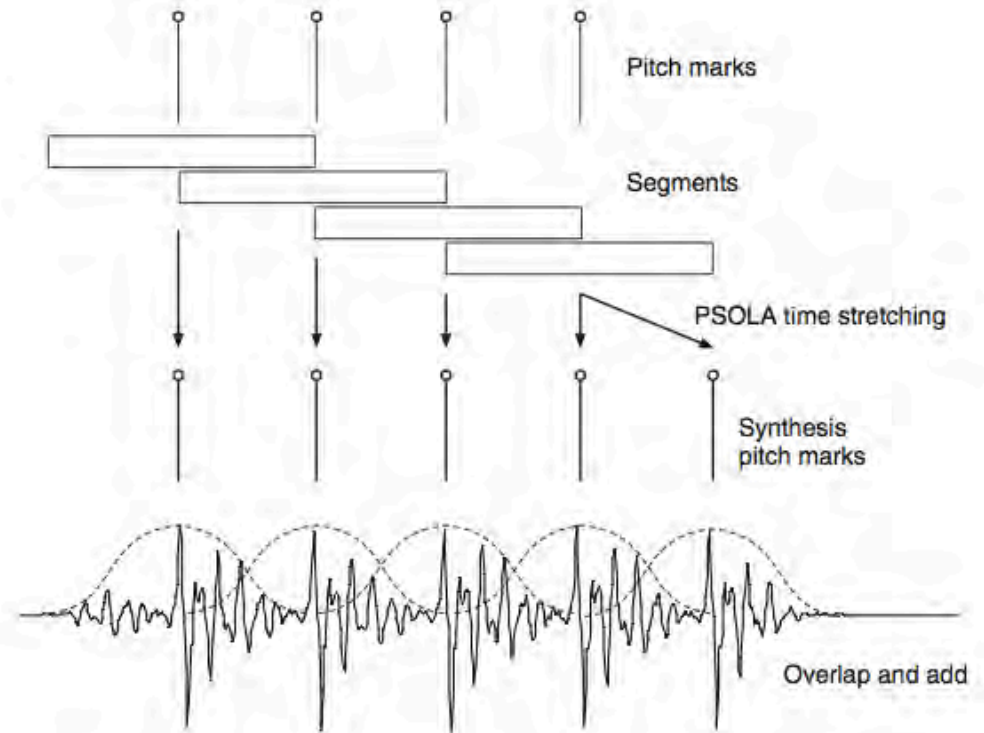
Pitch Synchronous Overlap and Add (PSOLA)

- Same idea as SOLA, but consider and preserve the local pitch
[Moulines et al., 1989, 1990]
- Developed specifically for speech signals.
- The algorithm is decomposed into two phases
 - Analysis: analyze and segment input
 - Synthesis: generate output by shifting and overlap-adding segments from analysis

PSOLA Graphically



(a) Analysis



(b) Synthesis

Figure from [Dutilleux et al. 2011]

PSOLA Analysis

- Determine the pitch period $P(t)$ and pitch marks t_i
 - Pitched parts
 - t_i spaced in time at the pitch rate
 - t_i should correspond to glottal pulse
 - Un-pitched parts
 - t_i spaced at a constant rate
- Extract segments centered at every pitch mark t_i , using a Hanning window of length $L_i = 2P(t_i)$ (two pitch periods).

PSOLA Synthesis

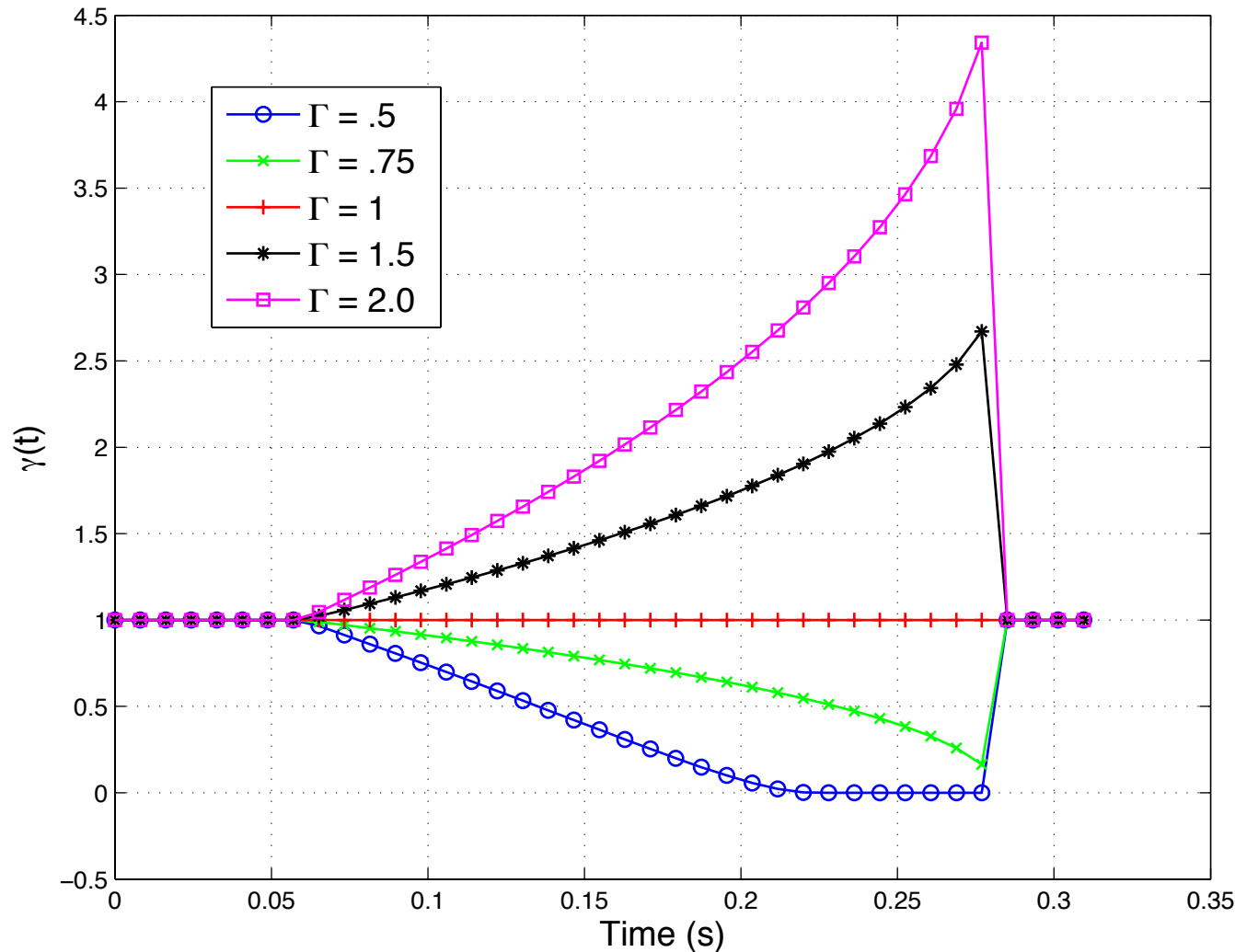
1. Choice of corresponding analysis segment i minimizing the time distance $|\alpha t_i - \tilde{t}_k|$
2. Overlap and add the selected segment: repeated for $\alpha > 1$ or discarded when $\alpha < 1$
3. Determine the time $\tilde{t}_{k+1}$ where the next synthesis segment will be centered (preserving local pitch)

$$\tilde{t}_{k+1} = \tilde{t}_k + \tilde{P}(\tilde{t}_k) = \tilde{t}_k + P(t_i)$$

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Effect of Stretch Length



$$\begin{aligned} &\underset{x}{\text{minimize}} && \|f\|_2 + \mu \|x\|_2 \\ &\text{subject to} && (x + x_0)^T \mathbf{1} = L \\ &&& x + x_0 \geq 0 \end{aligned}$$

 $\Gamma = .5$

 $\Gamma = .75$

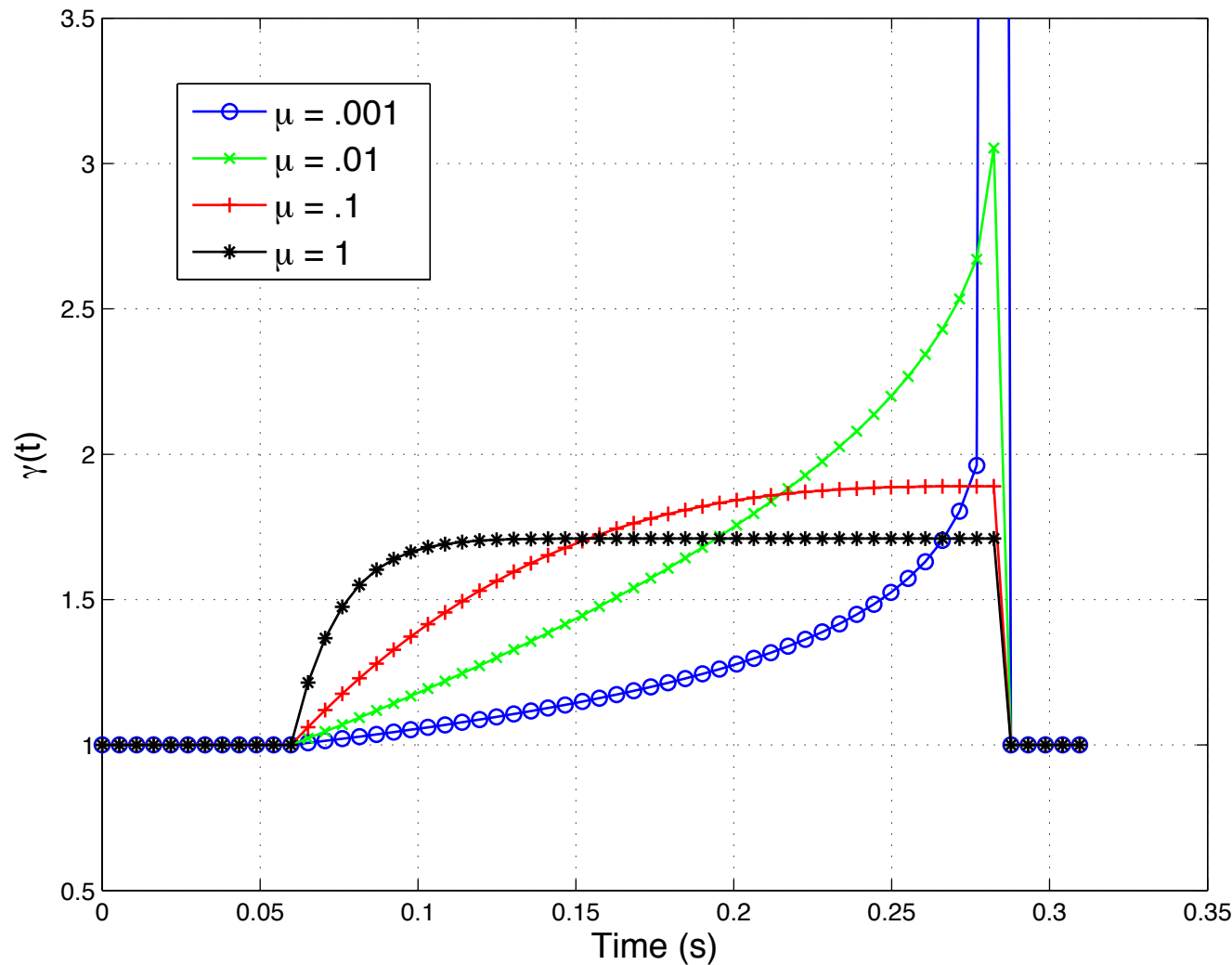
 $\Gamma = 1.0$

 $\Gamma = 1.5$

 $\Gamma = 2.0$

- Varying the overall stretch factor gives smooth, intuitive stretch factors

Effect of Regularization



$$\begin{aligned} &\underset{x}{\text{minimize}} && \|f\|_2 + \mu \|x\|_2 \\ &\text{subject to} && (x + x_0)^T \mathbf{1} = L \\ &&& x + x_0 \geq 0 \end{aligned}$$

 $\mu = .001$

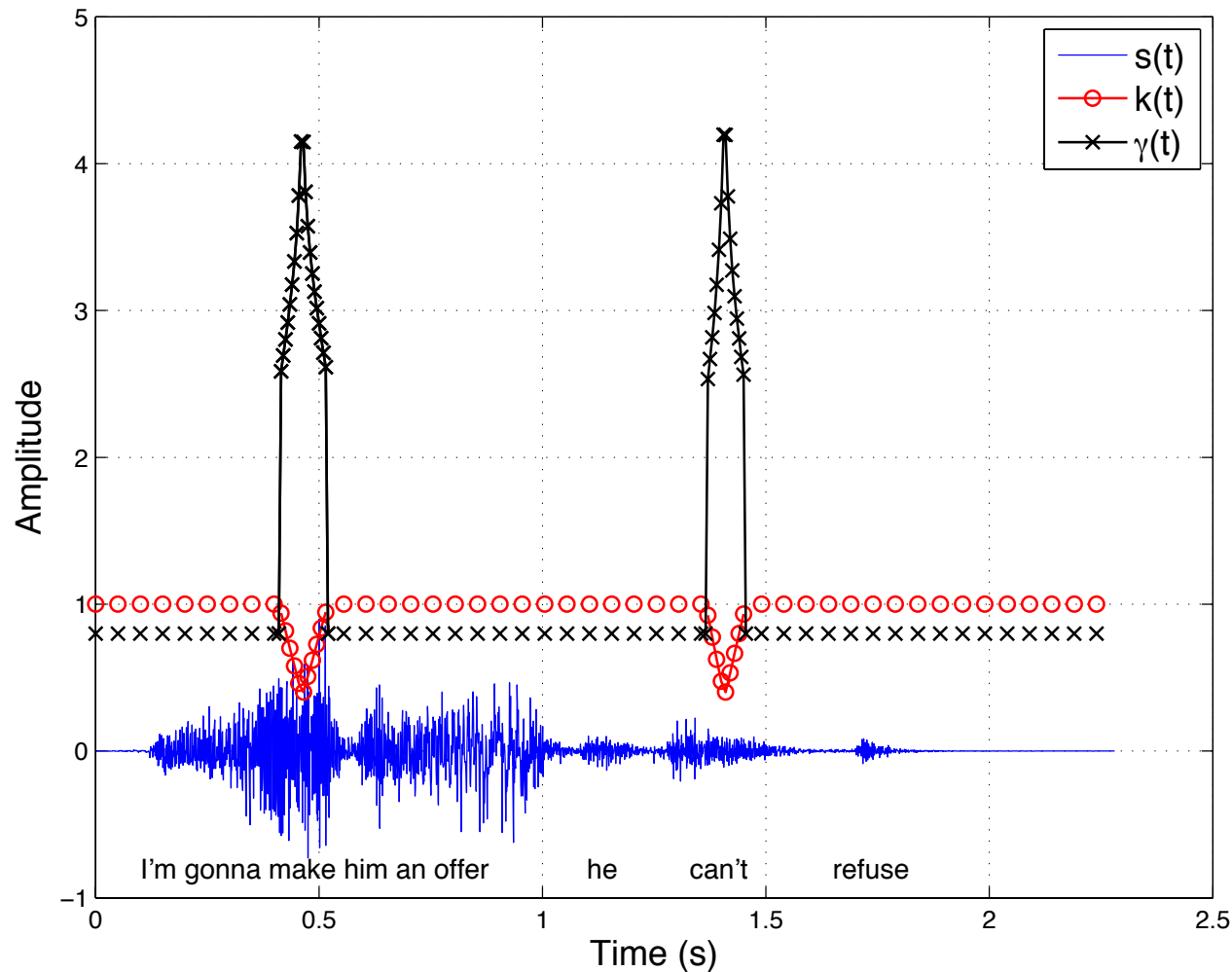
 $\mu = .01$

 $\mu = .1$

 $\mu = 1$

- Regularization penalty smooths the time-varying stretch factor

Demo - Emphasis Modification



 original

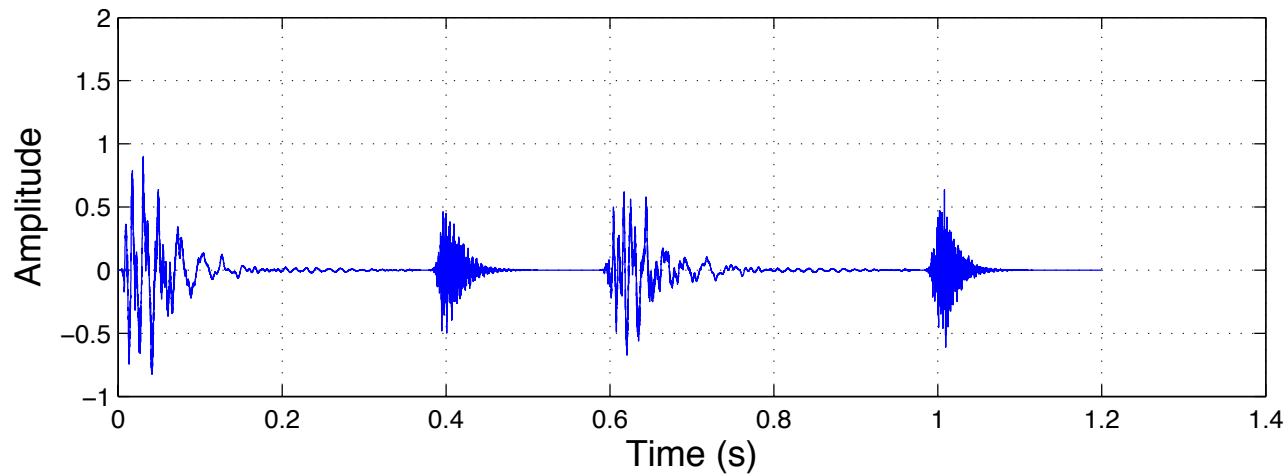
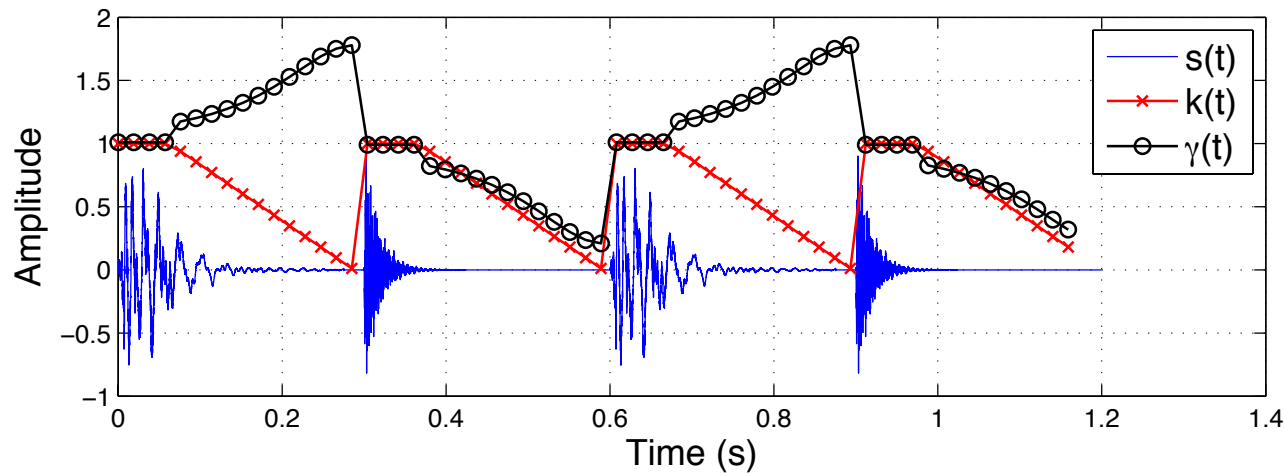
 warped

 warped + stretched
(slowed by 1.3x)

“I’m gonna make him an offer he can’t refuse.”

“I’m gonna maaake him an offer he caaaan’t refuse.”

Demo - Rhythmic Warping



 Original

No Stiffness / Stiffness

 Stretched (2x) 

 Swung +
Stretched (2x) 

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Conclusions

- Method of **user control** over variable-rate time-stretching
- **Decouples** stiffness control + timing constraints to user
- Converts user control into **optimal** time-dependent stretch rate
- **Agnostic** to time modification algorithms

References

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- E. Moulines and F. Charpentier. “Pitch-Synchronous waveform processing techniques for text-to-speech synthesis using diphones.” Speech communication 9.5, 1990.
- P. Dutilleux, G. De Poli, A. von dem Knesebeck, and U. Zoelzer, “Time-segment processing,” in Dafx: Digital Audio Effects, Udo Zoelzer, Ed. Wiley & Sons, Inc., NY, NY, USA, 2nd edition, 2011.

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